
This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Display
Manufacturer: Agile Display Solutions
Firmware Version: N/A
Model(s): HBX70LVHa, Monitor 70 Sharp

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.08.0002-b001	1.4.2

Version History

Module Version	Date	Notes
1_0_1_0	4/17/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: InterfaceName.connectionCounter = 5
- DeviceID variable must be set accordingly. Default value is '1'. Device ID ranges from 1-25 and Broadcast.
Example: InterfaceName.DeviceID = '1'

Supported Classes and Examples

SerialClass
InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='HBX70LVHa')
SerialOverEthernetClass
InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='HBX70LVHa')
EthernetClass
InterfaceName = ModuleName.EthernetClass('192.168.254.254', 23, Model='HBX70LVHa')

Set Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value 'Fill' 'Letter' ¹	Value 'Pillar'	Value 'Native' ¹
# AspectRatio example InterfaceName.Set('AspectRatio', 'Fill')			
Command AudioMute	Value 'Off'	Value 'On'	
# AudioMute example InterfaceName.Set('AudioMute', 'Off')			
Command AutoAdjust ³	Value None		
# AutoAdjust example InterfaceName.Set('AutoAdjust', None)			
Command Freeze ¹	Value None		
# Freeze example InterfaceName.Set('Freeze', None)			
Command Input	Value 'VGA' 'Displayport'	Value 'Digital DVI' 'HDMI 2' ¹	Value 'HDMI 1'
# Input example InterfaceName.Set('Input', 'VGA')			
Command MenuNavigation	Value 'Menu' 'Left' 'Exit'	Value 'Up' 'Right'	Value 'Down' 'Enter'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu')			
Command PIPAdjust ¹	Value 'Off' 'Large'	Value 'Small' 'Side-by-Side'	Value 'Medium'
# PIPAdjust example InterfaceName.Set('PIPAdjust', 'Off')			
Command	Value	Value	Value

PIPIInput ¹	'VGA' 'HDMI 2'	'Digital DVI' 'Displayport'	'HDMI 1'
# PIPInput example InterfaceName.Set('PIPIInput', 'VGA')			
Command PIPPosition ¹	Value 'Bottom-Left' 'Top-Right'	Value 'Bottom-Right'	Value 'Top-Left'
# PIPPosition example InterfaceName.Set('PIPPosition', 'Bottom-Left')			
Command PIPSwap ¹	Value None		
# PIPSwap example InterfaceName.Set('PIPSwap', None)			
Command Power	Value 'Off'	Value 'On'	
# Power example InterfaceName.Set('Power', 'Off')			
Command SchemeSelection	Value 'User' 'Cinema' ²	Value 'Sport' 'Vivid'	Value 'Game'
# SchemeSelection example InterfaceName.Set('SchemeSelection', 'User')			
Command Volume	Value 0 to 100 in steps of 1		
# Volume example InterfaceName.Set('Volume', 100)			
Command Zoom ¹	Value 'Zoom In'	Value 'Zoom Out'	
# Zoom example InterfaceName.Set('Zoom', 'Zoom In')			

¹ Supported by HBX70LVHa only.

² Does not work on VGA input.

³ Only works on VGA input.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value 'Fill' 'Letter' ¹	Value 'Pillar'	Value 'Native' ¹
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AudioMute	Value 'Off'	Value 'On'	
# AudioMute examples InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command Input	Value 'VGA' 'Displayport'	Value 'Digital DVI' 'HDMI 2' ¹	Value 'HDMI 1'
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command PIPAdjust ¹	Value 'Off' 'Large'	Value 'Small' 'Side-by-Side'	Value 'Medium'

<pre># PIPAdjust examples InterfaceName.Update('PIPAdjust') Value = InterfaceName.ReadStatus('PIPAdjust') InterfaceName.SubscribeStatus('PIPAdjust', None, FeedbackHandler)</pre>			
Command PIPIInput ¹	Value 'VGA' 'HDMI 2'	Value 'Digital DVI' 'Displayport'	Value 'HDMI 1'
<pre># PIPInput examples InterfaceName.Update('PIPInput') Value = InterfaceName.ReadStatus('PIPInput') InterfaceName.SubscribeStatus('PIPInput', None, FeedbackHandler)</pre>			
Command PIPPosition ¹	Value 'Bottom-Left' 'Top-Right'	Value 'Bottom-Right'	Value 'Top-Left'
<pre># PIPPosition examples InterfaceName.Update('PIPPosition') Value = InterfaceName.ReadStatus('PIPPosition') InterfaceName.SubscribeStatus('PIPPosition', None, FeedbackHandler)</pre>			
Command Power	Value 'Off'	Value 'On'	
<pre># Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>			
Command SchemeSelection	Value 'User' 'Cinema' ²	Value 'Sport' 'Vivid'	Value 'Game'
<pre># SchemeSelection examples InterfaceName.Update('SchemeSelection') Value = InterfaceName.ReadStatus('SchemeSelection') InterfaceName.SubscribeStatus('SchemeSelection', None, FeedbackHandler)</pre>			
Command Volume	Value 0 to 100 in steps of 1		
<pre># Volume examples InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)</pre>			

¹ Supported by HBX70LVHa only.

² Does not work on VGA input.

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 115200

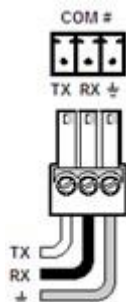
Data Bits: 8

Parity: None

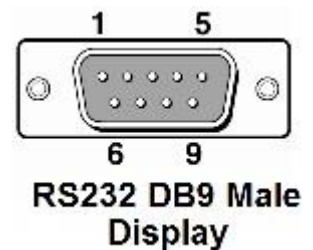
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	23
Logon Credentials Supported:	No
Multi-Connection	Undetermined
Capabilities:	
Port Changeability:	Undetermined

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Aspect Ratio Fill	\x07\x01\x02ASP\x01\x08
Aspect Ratio Letter	\x07\x01\x02ASP\x03\x08
Aspect Ratio Native	\x07\x01\x02ASP\x00\x08
Aspect Ratio Pillar	\x07\x01\x02ASP\x02\x08
Audio Mute Off	\x07\x01\x02MUT\x00\x08
Audio Mute On	\x07\x01\x02MUT\x01\x08
Auto Adjust None	\x07\x01\x02ADJ\x00\x08
Freeze None	\x07\x01\x02RCU\x18\x08
Input Digital DVI	\x07\x01\x02MIN\x01\x08
Input Displayport	\x07\x01\x02MIN\x0D\x08
Input HDMI 1	\x07\x01\x02MIN\x09\x08
Input HDMI 2	\x07\x01\x02MIN\x0A\x08
Input VGA	\x07\x01\x02MIN\x00\x08
Menu Navigation Down	\x07\x01\x02RCU\x03\x08
Menu Navigation Enter	\x07\x01\x02RCU\x06\x08
Menu Navigation Exit	\x07\x01\x02RCU\x07\x08
Menu Navigation Left	\x07\x01\x02RCU\x04\x08
Menu Navigation Menu	\x07\x01\x02RCU\x00\x08
Menu Navigation Right	\x07\x01\x02RCU\x05\x08
Menu Navigation Up	\x07\x01\x02RCU\x02\x08
PIP Adjust Large	\x07\x01\x02PSC\x03\x08
PIP Adjust Medium	\x07\x01\x02PSC\x02\x08
PIP Adjust Off	\x07\x01\x02PSC\x00\x08
PIP Adjust Side-by-Side	\x07\x01\x02PSC\x04\x08
PIP Adjust Small	\x07\x01\x02PSC\x01\x08
PIP Input Digital DVI	\x07\x01\x02PIN\x01\x08
PIP Input Displayport	\x07\x01\x02PIN\x0D\x08
PIP Input HDMI 1	\x07\x01\x02PIN\x09\x08
PIP Input HDMI 2	\x07\x01\x02PIN\x0A\x08
PIP Input VGA	\x07\x01\x02PIN\x00\x08
PIP Position Bottom-Left	\x07\x01\x02PPO\x00\x08
PIP Position Bottom-Right	\x07\x01\x02PPO\x01\x08
PIP Position Top-Left	\x07\x01\x02PPO\x02\x08
PIP Position Top-Right	\x07\x01\x02PPO\x03\x08
PIP Swap None	\x07\x01\x02SWA\x00\x08
Power Off	\x07\x01\x02POW\x00\x08

Global Scripter Module Communication Sheet

Revision: 4/17/2018

Power On	\x07\x01\x02POW\x01\x08
Scheme Selection Cinema	\x07\x01\x02SCM\x03\x08
Scheme Selection Game	\x07\x01\x02SCM\x02\x08
Scheme Selection Sport	\x07\x01\x02SCM\x01\x08
Scheme Selection User	\x07\x01\x02SCM\x00\x08
Scheme Selection Vivid	\x07\x01\x02SCM\x04\x08
Volume 0	\x07\x01\x02VOL\x00\x08
Volume 100	\x07\x01\x02VOLd\x08
Zoom Zoom In	\x07\x01\x02ZOM\x00\x08
Zoom Zoom Out	\x07\x01\x02ZOM\x01\x08

Appendix B. Update Commands

Aspect Ratio	\x07\x01\x01ASP\x08
Audio Mute	\x07\x01\x01MUT\x08
Input	\x07\x01\x01MIN\x08
PIP Adjust	\x07\x01\x01PSC\x08
PIP Input	\x07\x01\x01PIN\x08
PIP Position	\x07\x01\x01PPO\x08
Power	\x07\x01\x01POW\x08
Volume	\x07\x01\x01VOL\x08